Vincent Wren

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Education

University of Houston **May 2027**

Honors B.S. in Computer Science — Minors in Architecture, Data & Society *GPA: 3.69*

- **Honors:** Provost Undergraduate Research Scholarship; Houston Livestock Show and Rodeo Scholar
- **Relevant Coursework:** History of the Built Environment, Space & Movement in the City, Data Structures, Algorithms

Vrije Universiteit Amsterdam **Jan. 2025 – Jun. 2025**

Computer Science Exchange Student

- **Relevant Coursework:** Software Development, Machine Learning

Experience

Farm&City (Vision Zero Texas) **Sep. 2026 – Present**

Crash Data and Mapping Intern *Austin, TX (Remote)*

- Ranked Texas's most dangerous roads statewide, a safety counterpart to TxDOT's Top 100 Congested Roadways list.
- Screened 2.76M crash records on 82,000 miles of road with HSM network screening, ranking 5,960 corridors by crash cost.
- Automated 181 print-ready map pages, one for each Texas House and Senate district, with ArcGIS Pro and arcpy.
- Found that 10% of major-road miles carry 61% of the state's traffic fatalities and serious injuries.
- Presented the findings to staff from **15 legislative offices** at a Texas Capitol traffic safety briefing.

Pharis Fellowship **May 2026 – Aug. 2026**

Research Fellow, UH Honors College & HPE Data Science Institute *Houston, TX*

- Built and maintain **Vision Zero Houston**, a public dashboard of 421,000+ crashes, with Council Member Panzarella.
- Quantified crash concentration: half of Houston's traffic deaths and serious injuries occur on 2% of street-miles.
- Developed a street-design risk model capturing 51% of held-out severe crashes vs. 46% for the City's High Injury Network.
- Identified ∼300 miles of high-risk streets missing from the City's High Injury Network.

Lazarus Energy **Oct. 2023 – May 2026**

Graduate Intern *Houston, TX*

- Led 36 students through a year-long human-centered design program as teams built inventions for real users.
- Earned the **Project Invent Pioneer Award** for program leadership; student teams won $1,500 in prizes.

City of Houston **Jun. 2024 – Aug. 2024**

TechConnect Intern *Houston, TX*

- Led STEM activities for youth at TechConnect Fairs to promote digital equity; earned Apple Teacher certification.

Research

Wren, V. "Quantifying What Houston's High Injury Network Misses with a Validated Design-Risk Model." *Transportation Research Board 106th Annual Meeting*, Washington, D.C., Jan. 2027. **Accepted for presentation; sole author.**

Projects

Equitable Bike Share Siting Methodology | *Provost Undergraduate Research Scholarship* **Spring 2026**

- Designed a location-allocation model reaching 197,385 residents in a 5-minute walk, **2.65x the 2021 BCycle system**.
- Developed an equity score (poverty, race, vehicle access) to shift stations toward underserved neighborhoods.

Houston Transit Equity & Frequency Study | *ArcGIS Pro, GTFS* **Fall 2025**

- Mapped METRO's frequent network (GTFS) with 0.25-mile walksheds over Block Group population to find access gaps.

Downtown Houston Land Use Analysis | *ArcGIS Pro, OpenStreetMap* **Fall 2025**

- Classified downtown surface area by use: 71.9% of land serves vehicles vs. 9.2% for pedestrians.

Midtown Complete Streets Visualization | *ArcGIS CityEngine* **Fall 2025**

- Modeled a 3D road diet of Louisiana & Smith, turning five car lanes into sidewalks, a protected bike lane, and a bus lane.

Houston Equity of Places | *QGIS* **Summer 2025**

- Joined amenities to Census data: majority-Black neighborhoods hold 7.9% of residents but 1.3% of grocery stores.

Skills

GIS: ArcGIS Pro (map series, arcpy), Network Analyst, ArcGIS Online, CityEngine, QGIS, Kepler.gl
Methods: Network screening (HSM), crash cost analysis, location-allocation, regression modeling, georeferencing
Data & Programming: Python (pandas, matplotlib), JavaScript, Git/GitHub; TxDOT CRIS, Census, OSM, GTFS
Languages: English (Native), Dutch (Intermediate)

Leadership & Volunteering

CEA CAPA Alumni Ambassador | *Featured national storyteller on Dutch cycling infrastructure* **Feb. 2026**
No Higher No Wider I-10 | *Website Manager for the campaign against I-10 expansion* **Feb. 2024 – Present**
BikeHouston | *GearShifter Advocate; helped build a dashboard of Houston's bike network* **Jan. 2024 – Present**
Gehl | *Volunteer field data collector for the Downtown Houston Public Life Survey* **Oct. 2024**